

researchers—for free—and expect a real-time response. Such a service (at scale) would lose money, but it is easy to build on a micro scale. At that scale, it allowed Max and Damon to answer these all-important questions: If we can solve the tough technical problems behind this artificial intelligence product, will people use it? Will their use lead to the creation of a product that has real value?

It was this system that allowed Max and Damon to pivot over and over again, rejecting concepts that seemed promising but that would not have been viable. When they were ready to start scaling, they had a ready-made road map of what to build. The result: Aardvark was acquired for a reported \$50 million—by Google.⁹

THE ROLE OF QUALITY AND DESIGN IN AN MVP

One of the most vexing aspects of the minimum viable product is the challenge it poses to traditional notions of quality. The best professionals and craftspersons alike aspire to build quality products; it is a point of pride.

Modern production processes rely on high quality as a way to boost efficiency. They operate using W. Edwards Deming's famous dictum that the customer is the most important part of the production process. This means that we must focus our energies exclusively on producing outcomes that the customer perceives as valuable. Allowing sloppy work into our process inevitably leads to excessive variation. Variation in process yields products of varying quality in the eyes of the customer that at best require rework and at worst lead to a lost customer. Most modern business and engineering philosophies focus on producing high-quality experiences for customers as a primary principle; it is the foundation of Six Sigma, lean manufacturing, design thinking, extreme programming, and the software craftsmanship movement.

These discussions of quality presuppose that the company already knows what attributes of the product the customer will perceive as worthwhile. In a startup, this is a risky assumption to make. Often